Ana III Zusammenfassung

Moritz

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1 Measure Theory

Definition 1.1 (outer measure)**.** Let X be any set and let $\mu : 2^X \to [0, \infty]$ then μ is a outer measure if

1. $\mu(\emptyset) = 0$
2. $\mu(A) \leq \sum_k \mu(A_k)$ for every $A \subset \bigcup_k A_k$

Definition 1.2 (measurability)**.** A set $A \subset X$ is considered μ-measurable if for every $B \subset X$ we have the inequality

$$\mu(B) \geq \mu(A \cap B) + \mu(B \backslash A)$$

Now let $\mathcal{M}(\mu)$ be the set of all μ-measurable sets.

Theorem 1.3 (basic properties of measurable sets)**.** *Let $A \subset X$ be measurable and let $A_k \subset X$ be a sequence of measurable sets then we have the following properties*

1. $X \backslash A = A^C \in \mathcal{M}(\mu)$
2. $\bigcap_k A_k \in \mathcal{M}(\mu)$ *and* $\bigcup_k A_k \in \mathcal{M}(\mu)$
3. *if A_k is a pairwise disjoint sequence then we have* $\mu(\bigcup_k A_k) = \sum_k \mu(A_k)$